

Formative Evaluation Plan

Healthcare_FormativeHygiene

Introduction & Background

Healthcare staff operate in fast-paced, high-pressure environments where phishing and social engineering attacks pose significant risks to patient data and hospital systems. Our instructional design project develops a 30-minute eLearning module that uses simulations, case studies, and performance support tools to help nurses and clinical staff:

- Recognize phishing and scam indicators.
- Apply secure communication protocols.
- Report cybersecurity threats using hospital procedures.
- Demonstrate accountability and vigilance in daily digital practices.

The course is designed for varied learner profiles (tech-comfortable, tech-anxious, and tech-savvy staff) and incorporates accessibility features such as closed captions, high-contrast visuals, and mobile responsiveness.

This formative evaluation plan ensures the course is refined during development to maximize clarity, engagement, and transfer of learning before full implementation.

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Purpose of the Evaluation

To ensure the course meets its instructional goals and supports diverse learners, this evaluation focuses on several key areas.

The formative evaluation will:

- Assess accuracy, relevance, and clarity of cybersecurity content.
- Determine whether instructional strategies (branching scenarios, drag-and-drop, case studies) support learning objectives.
- Identify usability issues in navigation, accessibility, and learner engagement.
- Validate the effectiveness of assessments and performance support tools.
- Provide actionable recommendations for revisions prior to launch.

Learners for Evaluation:

- Nurses and clinical support staff representing diverse technical backgrounds.
- Subject Matter Experts (SMEs) in cybersecurity and healthcare compliance.
- Instructional designers and hospital IT staff for technical review.

Evaluation Questions

The following questions guide the evaluation and reflect the instructional design priorities outlined in the course.

Category	Evaluation Questions
Content Accuracy & Alignment	- Does the module content accurately reflect phishing and scam threats in healthcare? - Are examples realistic and relevant to hospital workflows? - Does content align with hospital protocols and compliance standards?
Instructional Design Effectiveness	Do the instructional strategies (branching scenarios, drag-and-drop, case studies) support the learning objectives? Is pacing appropriate for a 30-minute module? Are activities scaffolded to support different learner personas (tech-comfortable, tech-anxious, tech-savvy)?
User Experience & Accessibility	Is navigation intuitive and consistent? Do learners find the module engaging and applicable to their work?
Assessment Effectiveness	Do the knowledge checks align with objectives? Does feedback help learners correct mistakes?
Performance Support Resources	Is the downloadable checklist practical for daily use? Are additional resources needed to support transfer of learning?

Methods

To gather meaningful feedback, the evaluation will be conducted in four structured phases.

Phase	Participants	Process	Data Collection Tools
Expert Review	3–4 SMEs (cybersecurity, nurse educators)	Review content/design for accuracy, clarity, compliance	Structured feedback forms
One-to-One Testing	3–5 nurses	Think-aloud completion, observation, interview	Observation checklist + post-interview
Small Group Pilot	6–8 nurses	Group completion, survey, focus group discussion	Survey + focus group protocol
Technical Testing	2–3 LMS administrators	Browser/device/accessibility testing	Error logs + checklist

Sample Instrumentation

Each phase uses tailored instruments to collect both quantitative and qualitative data.

Survey (Likert scale + open-ended):

- The phishing scenarios were realistic and relevant to my role.
- Navigation was intuitive and easy to follow.
- The checklist will be useful in my daily work.
- Accessibility features (captions, high-contrast visuals) supported my learning.

Observation Checklist:

- Time per section.
- Navigation errors.
- Visible confusion or disengagement.
- Use of accessibility features.

Focus Group Questions:

- Which activity helped you most in recognizing phishing attempts?
- What challenges did you face while completing the module?
- How confident are you in applying the reporting protocol after training?
- Would you use the checklist or manager toolkit in your daily workflow?

Limitations

While the evaluation is designed to be rigorous, several limitations may affect interpretation and generalizability.

- Small sample sizes may not represent all hospital staff.
- Artificial testing environments may differ from real-world pressures.
- Self-reported data may not fully reflect actual behavior change.
- Limited timeframe prevents measurement of long-term retention.
- Cultural and organizational differences across hospitals may affect generalizability.

Analysis

The collected data will be analyzed using a combination of statistical and thematic techniques.

Quantitative:

- Average survey ratings, pre/post-test scores, completion times.

Qualitative:

- Thematic coding of open-ended responses.
- Categorization of usability issues and suggestions.

Integrated:

- Matrix mapping findings to learning objectives.
- Prioritization of revisions based on frequency and severity of issues.

Results (Hypothetical)

Preliminary findings from the pilot evaluation suggest the following outcomes.

- Content rated 4.7/5 for accuracy and relevance.
- Learning gains of 38% between pre- and post-tests.
- Navigation rated 4.5/5, though some confusion noted with branching scenarios.
- 82% of learners reported confidence in applying reporting protocols.
- Checklist rated highly practical; 88% said they would use it on the job.
- Accessibility features (captions, high-contrast visuals) rated 4.6/5 for effectiveness.

Recommendations

Based on the results and instructional design feedback, the following recommendations are proposed.

High Priority:

- Simplify branching scenario navigation.
- Add more examples of emotional manipulation tactics.
- Strengthen reporting workflow guidance.

Medium Priority:

- Enhance feedback in drag-and-drop activities.

- Provide additional case studies tailored to different learner personas.
- Improve accessibility toggles for flexible learner needs.

Low Priority:

- Expand optional deep resources for advanced learners.
- Add enrichment modules for leadership roles.

Recommendations will be prioritized based on learner impact, feasibility, and available budget.

High-priority changes will be implemented before launch; medium and low-priority updates will be scheduled for future iterations.

Conclusion

This formative evaluation plan ensures that the Cyber Hygiene for Healthcare module is refined through expert review, learner feedback, and technical testing before full deployment. By systematically addressing content accuracy, instructional design, usability, accessibility, and assessment effectiveness, the evaluation will provide actionable insights to strengthen the training's impact on protecting patient data and hospital systems.